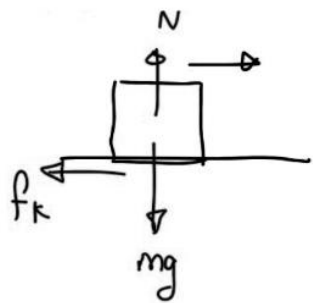


JEE-Main-05-04-2024 (Memory Based)
[EVENING SHIFT]

Physics

Question: A body of mass of 50 kg is moving on a rough horizontal surface having friction coefficient 0.3. Find the kinetic friction.



Options:

- (a) 147 N
- (b) 1.47 N
- (c) 14.7 N
- (d) 1470 N

Answer: (a)

Question: A body is moving in one direction with constant power. The displacement is proportional to time as

Options:

- (a) T
- (b) $T^{3/2}$
- (c) $T^{1/2}$
- (d) T^0

Answer: (b)

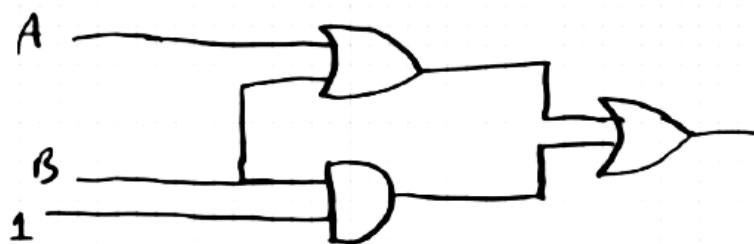
Question: In hydrogen spectrum, if shortest wavelength of Lyman series is $916 \text{ \AA}$ then find longest wavelength of Balmer series for hydrogen is

Options:

- (a) $95 \text{ \AA}$
- (b) $65 \text{ \AA}$
- (c) $9565 \text{ \AA}$
- (d) $6595 \text{ \AA}$

Answer: (d)

Question: Which combination of A & B gives zero as output?

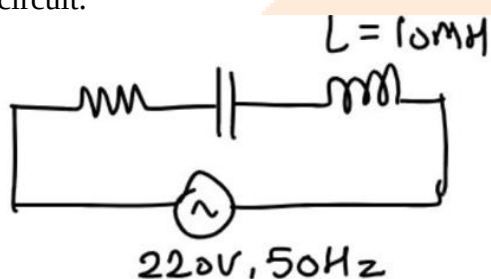


Options:

- (a) $A = 0, B = 0$
- (b) $A = 1, B = 0$
- (c) $A = 0, B = 1$
- (d) $A = 1, B = 1$

Answer: (a)

Question: For a series LCR circuit having AC source of 220V, 50Hz, It is observed that the voltage across the inductor which has self inductance of 10mH is 31.4 V. Find the current in circuit.



Options:

- (a) 10 mA
- (b) 10 A
- (c) 63 A
- (d) 63 mA

Answer: (b)

Question: If T is the time period of a satellite moving around earth in an orbit of radius R then find the time period of a satellite moving around another planet having mass as one fourth of that of earth and having same radius of orbit

Options:

- (a) $T/2$
- (b) $4T$
- (c) $2T$
- (d) $T/4$

Answer: (c)

Question: Which of the following statement/s about for stopping potential in a photoelectric effect is/are not correct.

Options:

- (a) Stopping potential depends on intensity light falling
- (b) Stopping potential is $1/e$ times max KE of photoelectrons
- (c) Stopping potential not constant for a given metal
- (d) Stopping potential is scalar physical quantity

Answer: (a)

Question: The equation for real gas is given by

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

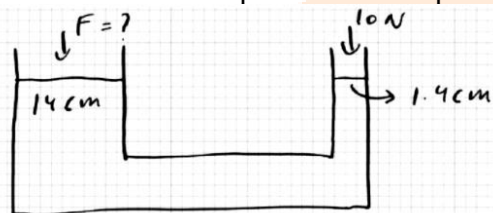
Find dimensions of a^1b^{-1} here, P is pressure, V is volume, T is temperature and R is gas constant.

Options:

- (a) $[M^1L^5T^{-2}]$
- (b) $[M^5L^2T^{-2}]$
- (c) $[M^1L^2T^{-2}]$
- (d) $[M^1L^2T^{-5}]$

Answer: (c)

Question: In the case of hydraulic lift 10 N force is applied to a side where the radius for the cross-section is 1.4 cm. The force that can be applied, on other side of containers having radius 14 cm to keep the fluid in equilibrium is given as 10n N find n



Options:

Answer: (3)

Question: A projectile is thrown with velocity v at an angle θ with horizontal, the maximum height up it goes is 64m. Keeping angle of projectile same, if initial velocity is halved then, find its maximum height.

Options:

- (a) 8 m
- (b) 16 m
- (c) 18 m
- (d) 32 m

Answer: (b)

Question: Truck carrying inflammable liquid has hanging metal chains why?

Options:

- (a) To alert others to take Caution
- (b) It is a Custom
- (c) To take care of the Tyre
- (d) To Neutralise free electrons due to air friction

Answer: (d)

Question: If a charged particle is moving in a region having uniform electric as well as uniform magnetic field. What is the electric force ($\vec{F}_1$) and the magnetic force ($\vec{F}_2$) acting on it are

Options:

- (a) $\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{V} \times \vec{E})$
- (b) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{B} \times \vec{V})$
- (c) $\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{E} \times \vec{V})$

(d) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V} \times \vec{B})$

Answer: (d)

Question: In the case of inductor the current is varying as $i = 3t + 8$. The induced e.m.f is 12V. Find the value of self inductance

Options:

- (a) 2 H
- (b) 4 H
- (c) 6 H
- (d) 8 H

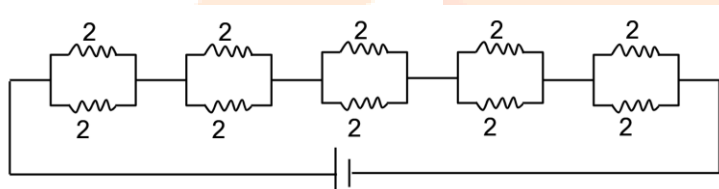
Answer: (b)

Question: A force acts on a body such that Momentum is $\vec{P} = \cos(kt) \hat{i} - \sin(kt) \hat{j}$. Find the Angle between $\vec{P}$ & $\vec{F}$

Options:

Answer: (90°)

Question: 10 Resistors of 1 Ohm Each arranged as shown. What will be the equivalent resistance



Options:

Answer: (5 ohm)

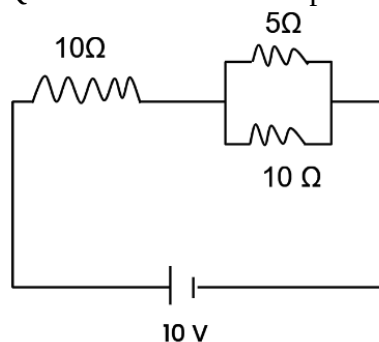
Question: In Adiabatic process Pressure is directly proportional to the cube of the temperature. Find γ

Options:

- (a) $3/2$
- (b) $4/3$
- (c) $7/3$
- (d) $7/5$

Answer: (a)

Question: Ratio of heat produced in 5 Ohm & 10 Ohm is



Options:

- (a) 1

- (b) 2
- (c) 3
- (d) 4

Answer: (b)

Question: If in a sonometer wire experiment if length of wire is 90 cm fundamental freq is 400Hz. If fundamental is 600 Hz what will be the length of wire

Options:

- (a) 30 Cm
- (b) 40 Cm
- (c) 50 Cm
- (d) 60 Cm

Answer: (d)

Question: Assertion: The light of Red colour deviates least from the prism as compared to violet and yellow colours.

Reason: Different colours deviate differently on passing through the prism depending on their wavelength

Options:

- (a) If both assertion and reason are true and the reason is the correct explanation of assertion
- (b) If both assertion and reason are true, but the reason is not the correct explanation of assertion
- (c) If the assertion is true, but the reason is false
- (d) If the assertion is false, but the reason is true

Answer: (b)

Question: The angular momentum of an electron in a hydrogen atom is proportional to (where r is radius of orbit)

Options:

- (a) r^1
- (b) r^{-1}
- (c) $r^{1/2}$
- (d) $r^{-1/2}$

Answer: (c)

Question: A galvanometer of resistance 100 ohm measure a maximum voltage of 10 volt when connected in series to a 400 ohm resistor. What resistance has to be connected in parallel to galvanometer to convert it into an ammeter of range 10 ampere

Options:

- (a) 1 Ω
- (b) 2 Ω
- (c) 3 Ω
- (d) 4 Ω

Answer: (a)

Question: The mean free path for a gas, with molecular diameter d and number density n can be expressed as:

Options:

- (a) $\frac{1}{\sqrt{2}n^2\pi d}$

(b) $\frac{1}{\sqrt{2}n^2\pi^2d^2}$

(c) $\frac{1}{\sqrt{2}n\pi d}$

(d) $\frac{1}{\sqrt{2}n\pi d^2}$

Answer: (d)



JEE-Main-05-04-2024 (Memory Based)
[EVENING SHIFT]

Chemistry

Question: How many can give H_2 gas from dil acid Ti^{2+} , Cr^{2+} , V^{2+}

Options:

Answer: (5)

Question: If shortest wavelength of Lyman series is 916A the longest wave of balmer series is

Options:

- (a) $\frac{9x}{5}$
- (b) $\frac{36x}{5}$
- (c) $\frac{x}{4}$
- (d) $\frac{5x}{9}$

Answer: (d)

Question: Mass of Ag deposited by 1 coulomb charge

Options:

- (a) 1 electrochemical equivalent
- (b) 1 g
- (c) .1 g
- (d) 1 chemical equivalent

Answer: (a)

Question: How many have dipole moment zero. HF , $BeCl_2$, BeF_2 , BF_3 , SiF_4 , NH_3 , H_2O , H_2S , NF_3 , CH_4 , $CHCl_3$, CO_2 , H_2

Options:

Answer: (6)

Question: Find out E cell of the given cell $M | M^{2+} || X^{2-} | X$.

$$E^\circ_{M^{2+} | M} = 0.34 \text{ V}$$

$$E^\circ_{X | X^{2-}} = 0.46 \text{ V}$$

Options:

- (a) 0.80 V
- (b) 0.12 V
- (c) -0.12 V
- (d) -0.80 V

Answer: (a)

Question: Which of the following is true regarding coagulation of egg :

Options:

- (a) 1° structure does not change
- (b) 2° structure does not change
- (c) 3° structure does not change
- (d) Denaturation of protein does not occur

Answer: (a)

Question: Angular momentum of an electron in an orbit of radius R of a hydrogen atom is directly proportional to

Options:

- (a) R
- (b) 1/R
- (c) $1/\sqrt{R}$
- (d) $\sqrt{R}$

Answer: (d)

Question: Assertion :- Dipole moment of NH_3 is greater than NF_3

Reason :- Dipole moment of N-H aligns with the dipole moment of N and lone pair. F has high electronegativity

Options:

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: (a)

Question: Find out value of C_p / C_v for an ideal gas undergoing reversible adiabatic process for which $P \propto T^3$ is given

Options:

- (a) 4/3
- (b) 3/2
- (c) 4/5
- (d) 5/4

Answer: (b)

Question: $M | M^{2+} || X | X^{2-}$

$E_{M^{2+}}$, $m = 0.46$, E_X , $X^{2-} = 0.34$

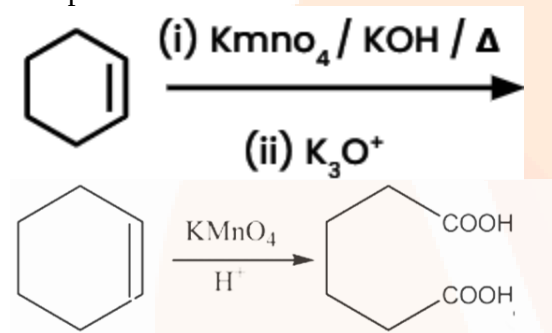
Options:

- (a) rxn $m + x \rightarrow m^{2+} x^{2-}$ is spontaneous
- (b) rxn $m^{2+} + x^{2-} \rightarrow m + x$ is spontaneous
- (c) e cell of rxn according to que=some value
- (d) This was also like option D

Answer: (b)

Question: Consider the following reaction :

The product is



Options:

- (a) Adipic Acid
- (b) Oxalic Acid
- (c) Succinic Acid
- (d) Benzoic Acid

Answer: (a)

Question: In an atom, how many electrons can have

(i) $n = 4$

(ii) $ml = 1$

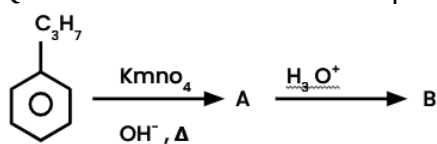
(iii) $ms = 1/2$

Options:

- (a) 32
- (b) 16
- (c) 8
- (d) 2

Answer: (b)

Question: Number of π bonds present in product B is :



Options:

- (a) Benzoic Acid
- (b) Adipic Acid
- (c) Succinic acid
- (d) None of the above

Answer: (a)

Question: For the reaction $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

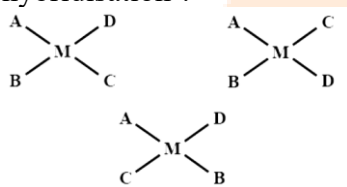
How many moles of methane is required for the formation of 11 g of CO_2 .

Options:

- (a) 0.3
- (b) 0.25
- (c) 0.5
- (d) 2

Answer: (b)

Question: Number of geometrical isomerism possible for Mabcd type compound having sp^3 hybridisation ?



Options:

- (a) 3
- (b) 6
- (c) 4
- (d) 5

Answer: (a)

Question: Number of correct statements ?

Statement - 1 : In group 13, atomic radius increases down the group.

Statement - 2 : Every element of group 13 have stable stable +1 oxidation state.

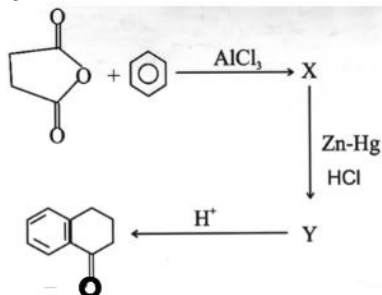
Statement - 3 : For group 13 element electronegativity decreases down the group.

Statement - 4 : Hybridisation of $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ is sp^3d^2 .

Statement - 5 : Aluminum is rendered passive by conc. HNO_3

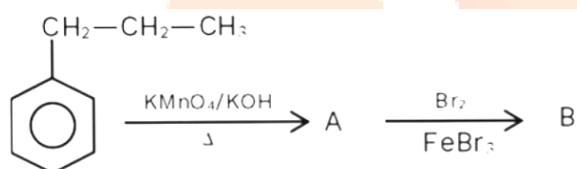
Options:
Answer: (3)

Question:

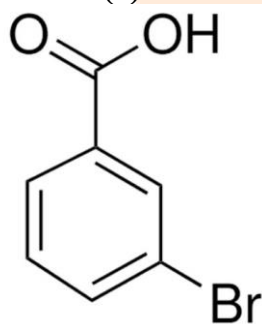


Options:
Answer: (z)

Question: Calculate number of π bond present in product B



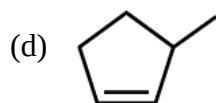
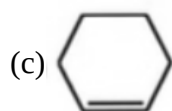
Options:
Answer: (8)



Question: Major product in the given reaction

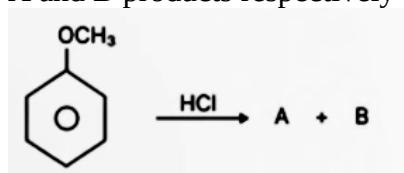
Options:



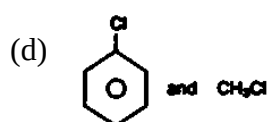
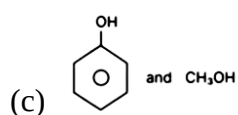
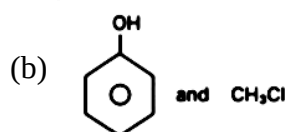
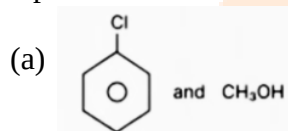


Answer: (d)

Question: Consider the following sequence of reaction :
A and B products respectively are :

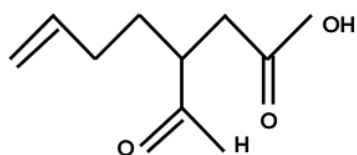


Options:



Answer: (b)

Question: What is the IUPAC of



Options:

(a) 3 - formylhept - 6 - enoic acid

(b) 3 - aldohept - 7 - enoic acid

(c) 3 - ketohept - 6 - enoic acid

(d) 3 - oxohept - 6 - enoic acid

Answer: (a)



JEE-Main-05-04-2024 (Memory Based)

[EVENING SHIFT]

Maths

Question: If $S = (2, 4, 8, \dots, 512)$ divided in 3 set A, B, C into equal no. of elements the $n(A \cup B \cup C)$ is equal to S then $A \cap B = B \cap C = C \cap A = \phi$ then probability of ABC

Question: The 50th word in the dictionary using the letters B, B, H, J, O is:

Options:

- (a) OBBJH
- (b) OBBHJ
- (c) JHBBO
- (d) BBHOJ

Answer: (a)

Question: $\left(\frac{3^{\frac{1}{5}}}{x} + \frac{2x}{5^{\frac{1}{3}}}\right)^{12}$. Find which term is constant.

Options:

- (a) 4th
- (b) 5th
- (c) 6th
- (d) 7th

Answer: (d)

Question: $4^{1+x} + 4^{1-x}, \frac{k}{2}, 16^x + 16^{-x}$ Let _____ are in A.P. then least value of k is _____.

Answer: (10)

Question: Let image of point (8, 5, 7) with respect to line

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5} \text{ is } (\alpha, \beta, \gamma). \text{ Then, } \alpha + \beta + \gamma \text{ is equal to}$$

Options:

- (a) 10
- (b) 12
- (c) 9
- (d) 14

Answer: (d)

Question: Area bounded by $y = -2|x|$ and $y = x|x|$ is:

Options:

- (a) $\frac{2}{3}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{2}$
- (d) $\frac{4}{3}$

Answer: (d)

Question: The number of real solution $x|x+5| + 2|x+7| - 2 = 0$ is

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (c)

Question: $A = \begin{bmatrix} \alpha & \alpha & \alpha \\ \beta & \alpha & -\beta \\ -\alpha & \alpha & \alpha \end{bmatrix}$. B is formed by cofactor of A, matrix then find determinant of AB.

Options:

- (a) $4\alpha^3(2\alpha + \beta)^5$
- (b) $12\alpha^3(2\alpha + \beta)^2$
- (c) $8\alpha^6(\alpha + \beta)^3$
- (d) $18\alpha^8(\alpha + \beta)^3$

Answer: (c)

Question: Consider the equation $P(x) = ax^2 + bx + c = 0$. If $a, b, c \in A$ where $A = \{1, 2, 3, 4, 5, 6\}$, then the probability that $P(x)$ has real and distinct roots?

Question: If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined such that $f(x) = |x| - 1$ and

$$g(x) = \begin{cases} e^x, & x > 0 \\ x - 1, & x \leq 0 \end{cases} \text{ then,}$$

Options:

- (a) Both f and g are one-one
- (b) f is one-one and g is many one
- (c) f is many-one and g is one-one
- (d) Both f and g are many-one

Answer: (c)

Question: A line L is perpendicular to $y = 2x + 10$ such that it touches the parabola $y^2 = 4(x - 9)$. Find distance between point of contact and origin.

Options:

- $\sqrt{165}$ (a)
- $\sqrt{175}$ (b)
- $\sqrt{185}$ (c)
- $\sqrt{190}$ (d)

Answer: (c)

Question: If $S = \{2, 3, 8, 16, \dots, 512\}$. If S is broke in 3 equal subsets A, B and C such that $A \cap B = B \cap C = C \cap A = \phi$ and $A \cup B \cup C = S$, then maximum number of ways to break is:

Options:

(a) 9C_3

(b) $\frac{9!}{(3!)^3}$

(c) $\frac{9!}{(3!)^4}$

(d) $\frac{9!}{(3!)^2}$

Answer: (b)

Question: If $\frac{dy}{dx} + y + y \cdot \frac{2x}{(1+x^2)^2} = xe^{\frac{1}{1+x^2}} \cdot y(0) = 0$, and $f(x) = y(x)e^{-\frac{1}{1+x^2}}$ and $y = x$.

Then the area bounded b/w these two curve equals to:

Options:

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) $\frac{7}{6}$

(d) 2

Answer: (a)

Question: If $y = \frac{2 \cos 2\theta + \cos \theta}{\cos 3\theta + \cos^2 \theta + \cos \theta}$, then the value of $y'' + y' + y$ is

Options:

(a) $\sec \theta (1 - \tan^3 \theta)$

(b) $\tan \theta (\sec^3 \theta + 2 \tan^2 \theta)$

(c) $\sec \theta (2 \sec^2 \theta + \tan \theta)$

(d) $\cot \theta (\sec^3 \theta + 2 \tan^2 \theta)$

Answer: (c)

Question: If $2x^2 - x + 2 = 0$ and one root is a, then $\lim_{x \rightarrow \frac{1}{a}} \frac{16(1 - \cos(2x^2 - x + 2))}{(ax - 1)^2}$ equal to

Options:

$\frac{32(1 - a^2)^2}{a^4}$ (a)

$\frac{8(1 - a^2)^2}{a^4}$ (b)

$\frac{16(1 - a^2)^2}{a^4}$ (c)

$\frac{20(1 - a^2)^2}{a^3}$ (d)

Answer: (a)

Question: $\beta(m, n) = \int_0^1 x^m (1 - x^m)^{n-1} dx$ and

$a \times \beta(-b, c) = \int_0^1 (1 - x^{10})^{20} dx$ then $(a + b + c)$ is equal to:

Options:

(a) 210

(b) 230

(c) 250

(d) 270

Answer: (a)

Question: Find the differential equation of circle whose centre lies on $y = x$ and passes through $(0, 1)$

Options:

$$-x^2 + y^2 - 2xy + 2x - 1 + \frac{dy}{dx} \quad (\text{a})$$

$$(x^2 - y^2 - 2xy + 2y - 1) = 0$$

$$-x^2 - y^2 - 2xy + 2x - 1 + \frac{dy}{dx} \quad (\text{b})$$

$$(x^2 + y^2) = 0$$

$$-x^2 - y^2 - 2xy + 2x - 1 + \frac{dy}{dx} \quad (\text{c})$$

$$(x^2 - y^2) = 0$$

$$-x^2 + y^2 - 2xy + 2x - 1 + \frac{dy}{dx} \quad (\text{d})$$

$$(x^2 + y^2 - 2) = 0$$

Answer: (a)

Question: $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$, then minimum value of $|\vec{c} - \vec{a}|^2$

Options:

(a) 13

(b) 5

$$\frac{40}{9} \quad (\text{c})$$

$$\frac{20}{9} \quad (\text{d})$$

Answer: (c)